

Autocovariance function for X_t

$$X_t = \sum_{j=-n}^n R_j e^{i\omega_j t}$$

Consider $h > 0$

$$X_{t+h} = \sum_{k=-n}^n R_k e^{i\omega_k (t+h)}$$

Then

$$\gamma(h) = \text{Cov}(X_t, X_{t+h})$$

$$= \mathbb{E}(X_t X_{t+h}) - \mathbb{E}(X_t) \mathbb{E}(X_{t+h})$$

$$= \mathbb{E}(X_t X_{t+h})$$

↑ because

$\mathbb{E}(X_t) = 0$ for all t

$$= \mathbb{E} \left[\left(\sum_{j=-n}^n R_j e^{i\omega_j t} \right) \left(\sum_{k=-n}^n R_k e^{i\omega_k (t+h)} \right) \right] \quad (*)$$

X_t is a real-valued random variable

$\therefore X_t = \overline{X}_t \leftarrow$ complex conjugate of X_t

$$\overline{X}_{t+h} = \sum_{k=-n}^n \overline{R}_k e^{-i\omega_k (t+h)}$$

$$= X_{t+h}$$

$\therefore$ In $(*)$:

$$\gamma(h) = \mathbb{E} \left[\left(\sum_{j=-n}^n R_j e^{i\omega_j t} \right) \left(\sum_{k=-n}^n \overline{R}_k e^{-i\omega_k (t+h)} \right) \right] \quad (*)$$

$$\begin{aligned}\mathbb{E}(R_j \bar{R}_k) &= \mathbb{E}\left[\frac{1}{2}(P_j - iQ_j) \frac{1}{2}(P_k + iQ_k)\right] \\ &= 0 \quad \text{if } j \neq k.\end{aligned}$$

$\therefore$ The only non-zero terms in (*) will occur where $j=k$.

$\therefore$ (*) becomes

$$\gamma(h) = \mathbb{E}\left[\sum_{j=-n}^n R_j \bar{R}_j e^{i\omega_j t} e^{-i\omega_j(t+h)}\right]$$

$$= \mathbb{E}\left[\sum_{j=-n}^n R_j \bar{R}_j e^{-i\omega_j h}\right]$$

$$= \sum_{j=-n}^n \mathbb{E}(R_j \bar{R}_j) e^{-i\omega_j h}$$

$$= \sum_{j=-n}^n g_j e^{-i\omega_j h}$$

$$\gamma(0) = \text{Var}(X_t)$$

$$= \sum_{j=-n}^n g_j e^0$$

$$= \sum_{j=-n}^n g_j$$

g_j = contribution to the variance of the process $\{X_t\}$ from the frequency ω_j